
Online Examination Report

Date:	25.07.2014
Attempts:	1
Examinee:	Ojule, Benjamin
Date of birth:	
Subject:	020-AGK
Result:	32,49 %
Time used:	00:01:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	3648	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	2434	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
3	2405	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
4	279	21.3	HYDRAULICS	0,00	1,00
5	1233	21.3.2	Hydraulic systems	0,00	1,00
6	2367	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	1,00	1,00
7	556	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
8	212	21.8	FUEL SYSTEM	0,00	1,00
9	3	21.9	ELECTRICS	0,00	1,00
10	443	21.9.1	General, definitions, basic applications: etc	0,00	1,00
11	2288	21.9.1	General, definitions, basic applications: etc	0,00	1,00
12	1384	21.9.1.8	Semiconductors and logic circuits:	0,00	1,00
13	643	21.10.1.2	Engine: design, operation etc	0,00	1,00
14	2438	21.10.10.1	Performance	0,00	1,00
15	2251	21.11.1	Basic principles	0,00	1,00
16	1727	21.11.1.1	Basic generation of thrust, the thrust formula	0,00	1,00
17	1164	21.11.2.4	Turbine	0,00	1,00
18	1977	21.11.3.1	Engine fuel system	0,00	1,00
19	2147	21.13	OXYGEN SYSTEMS	1,00	1,00
20	1874	21.13	OXYGEN SYSTEMS	0,00	1,00
21	4121	22.	INSTRUMENTATION	0,00	1,00
22	3090	22.	INSTRUMENTATION	0,00	1,00
23	3100	22.	INSTRUMENTATION	0,00	1,00
24	3085	22.	INSTRUMENTATION	1,00	1,00
25	972	22.2.1	Pressure measurement	1,00	1,00
26	119	22.2.1	Pressure measurement	0,00	1,00
27	1884	22.2.1.1	Definitions	0,00	1,00
28	792	22.2.1.1	Definitions	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	260	22.2.4	Altimeter	1,00	1,00
30	1938	22.2.4	Altimeter	1,00	1,00
31	1045	22.2.4	Altimeter	0,00	1,00
32	65	22.2.6	Airspeed indicator (ASI)	1,00	1,00
33	1320	22.2.7	Mach meter	0,00	1,00
34	1646	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
35	1800	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	1,00	1,00
36	1595	22.4	GYROSCOPIC INSTRUMENTS	0,00	1,00
37	1522	22.4.1	Gyroscope: basic principles	1,00	1,00
38	1461	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
39	409	22.6.3	Flight Director : design and operation.	1,00	1,00
40	1422	22.6.3	Flight Director : design and operation.	1,00	1,00
Total:32%				13,00	40,00

Attachment 2: List of result by topic

Licence

Date: 25.07.2014

Examinee: Ojule, Benjamin

Location: Uganda Civil Aviation Authority

Your Result: 13,00 from 40,00 Points (32,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	3,00	20,00	15,00
21.2. AIRFRAME	2	0,00	2,00	0,00
21.3. HYDRAULICS	2	0,00	2,00	0,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	1,00	1,00	100,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	1	1,00	1,00	100,00
21.8. FUEL SYSTEM	1	0,00	1,00	0,00
21.9. ELECTRICS	4	0,00	4,00	0,00
21.10. PISTON ENGINES	2	0,00	2,00	0,00
21.11. TURBINE ENGINES	4	0,00	4,00	0,00
21.13. OXYGEN SYSTEMS	2	1,00	2,00	50,00
22. INSTRUMENTATION	20	10,00	20,00	50,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	9	5,00	9,00	55,56
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	2	1,00	2,00	50,00
22.4. GYROSCOPIC INSTRUMENTS	3	1,00	3,00	33,33
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	2	2,00	2,00	100,00
Total	40	13,00	40,00	32,50

1

2-cycle engine thrust and fuel efficiency can be greatly compromised when

- ☒ **intake valve lifters fail to pressurize and provide adequate fuel to the combustion chamber.**
- ☐ **carbon deposits build up on exhaust valves.**
- ☐ **exhaust systems are installed that are not specifically tuned for an engine.**

2

Identify the type of stability if the aircraft attitude remains in the new position after the controls have been neutralized.

☐ Negative longitudinal static stability.

☐ Neutral longitudinal static stability.

☒ Neutral longitudinal dynamic stability.

3

Which is a purpose of ground spoilers?

- ☐ Aid in rolling an airplane into a turn.
- ☐ Reduce the wings' lift upon landing.
- ☒ Increase the rate of descent without gaining airspeed.

4

Hydraulic fluids :

- ☒ Cause high fire risk.
- ☐ Do not require special care.
- ☐ Are irritating to eyes and skin.
- ☐ Are irritating to eyes and skin and cause high fire risk.

5

The component that transforms the hydraulic pressure into a linear motion is called ...

☐ A Pressure regulator.

☒ A hydraulic pump.

☐ An actuator or jack.

☐ An accumulator.

6

"Nose wheel shimmy" may be described as :

- ☒ a possibly damaging vibration of the nose wheel when moving on the ground.
- ☐ the oscillatory movement of the nose wheel when extended prior to landing.
- ☐ aircraft vibration caused by the nose wheel upon extension of the gear.
- ☐ the amount of free movement of the nose wheel before steering takes effect.

7

"Conditioned" air is air that has :

- ☒ been controlled in respect of temperature and pressure.
- ☐ had the oxygen content increased.
- ☐ had the oxygen content reduced.
- ☐ had any moisture removed from it.

8

In a compensated capacitance type quantity indicating system, the contents gauge of a half-full fuel tank indicates a fuel mass of 8000 lb. If a temperature rise increased the volume of fuel by 5 %, the indicated fuel weight would :

☒ increase by 5 %.

☐ increase by 10 %.

☐ remain the same.

☐ decrease by 5 %.

9

Electrical bonding of an aircraft is used to:

1. protect the aircraft against lightning effects.
2. set the electrostatic potential of the aircraft to a value approximating to 0 volt.
3. prevent radio interference on radio communication systems.
4. set the different aircraft parts to a single potential.

The combination that regroups all of the correct statements is:

☐ 1, 2, 4

☒ 1, 3, 4

☐ 2, 3, 4

☒ 1, 2, 3

10

On modern transport aircraft, cockpit windows are protected against icing by :

- ☐ Anti-icing fluid.
- ☐ Electric heating.
- ☐ Rain repellent system.
- ☒ Vinyl coating.

11

"Frequency wild" in relation to a AC generation system means the generator :

- ☐ output frequency is too high.
- ☒ output frequency varies with engine speed.
- ☐ output frequency is too low.
- ☒ voltage regulator is out of adjustment.

12

A feeder fault on a direct current circuit results from a flux unbalance between the:

- ☐ generator and the series winding turn.
- ☐ shunt exciter and the series winding turn.
- ☒ voltage coil and the series winding.
- ☐ voltage coil and the series winding turn.

13

The crank assembly consists of

- ☒ Crankcase, crankshaft, connecting rods and pistons.
- ☐ propeller, crankshaft, pistons and connecting rods.
- ☐ crankshaft, connecting rods and pistons.
- ☐ crankshaft, camshaft, valves, valve springs and push rods.

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14

If an engine failure occurs at an altitude above single-engine ceiling, what airspeed should be maintained?

☐ V(XSE).

☐ V(YSE).

☒ V(MC).

15

Max. Exhaust Gas Temperature is theoretically associated with :

- ☐ Cruising mixture setting.
- ☒ Mass ratio of 1/15.
- ☐ Mixture ratio very close to idle cut-out.
- ☒ Full rich setting.

16

Regarding a jet engine:

- I. The maximum thrust decreases as the pressure altitude increases.
II. The specific fuel consumption is inversely proportional to pressure altitude, at constant TAS.

- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

17

The control of free turbine engines on turboprops, is accomplished by: - a propeller control lever used to select: 1 - propeller RPM 2 - turbine temperature 3 - turbine RPM - a fuel control lever used to select: 4 - propeller RPM 5 - torque 6 - turbine temperature
The combination which regroups all of the correct statements is :

☐ 1 - 3 - 5

☐ 2 - 4 - 5

☐ 1 - 5 - 6

☒ 3 - 4 - 6

18

Fuel flow information for gas turbine engines is:

- ☐ *measured at the outlet of the high pressure fuel pump.*
- ☐ *not used on aircraft.*
- ☐ *measured in the line between the fuel control unit and the engine burners.*
- ☒ *measured in the line between the booster pump and the engine.*

19

The purpose of the "Pressure Relief Valve" in a high pressure oxygen system is to :

- ☐ maximize the charging pressure of the system
- ☐ reduce pressure in the oxygen reservoir to a suitable manifold pressure for the regulator
- ☐ act as a manual shut-off valve
- ☒ relieve overpressure if the pressure reducing valve malfunctions

20

The oxygen masks have dropped down from the passengers service units. The oxygen flow starts :

- ☐ immediately
- ☐ after pulling the oxygen mask downwards
- ☒ After the system has been switched on by a crew member
- ☐ only above FL200

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21

(Refer to Figure 83.) Which altimeter depicts 12,000 feet?

See Attachments:

1. instrument_83.jpg

figure 83

(Attachment
No.1)

☒ 2

☐ 4

☐ 3

22

Identify taxi leadoff lights associated with the centerline lighting system.

- ☒ **Alternate green and yellow lights curving from the centerline of the runway to the centerline of the taxiway.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to a point on the exit.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to the edge of the taxiway.**

23

Identify the runway distance remaining markers.

- ☒ Signs with increments of 1,000 feet distance remaining.
- ☐ Yellow marker laterally placed across the runway with signs on the side denoting distance to end.
- ☒ Red markers laterally placed across the runway at 3,000 feet from the end.

24

Identify touchdown zone lighting (TDZL).

- ☒ Two rows of transverse light bars disposed symmetrically about the runway centerline.
- ☐ Alternate white and green centerline lights extending from 75 feet from the threshold through the touchdown zone.
- ☐ Flush centerline lights spaced at 50-foot intervals extending through the touchdown zone.

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25

If the static pressure ports iced over while descending from altitude, the airspeed indicator would read:

☐ Correctly.

☐ Low

☒ High

26

An aircraft is flying from a cold air mass into a warm air mass. The TAS and true altitude will:

☒ TAS increases, true altitude decreases.

☐ Both increases.

☐ TAS decreases, true altitude increases.

27

Concerning the pitot and static system, the static pressure error:

- ☐ *is a direct effect of heating of the static ports.*
- ☒ *is a direct effect of a blockage of the static port.*
- ☐ *affects the alternate static port only.*
- ☐ *is caused by disturbed airflow around the static ports.*

28

Given:

Pt = total pressure

Ps = static pressure

Pso = static pressure at sea level

The CAS is a function of:

☐ $(Pt - Pso) / Ps$

☒ $Pt - Ps$

☐ Pt / Ps

☐ $Pt - Pso$

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29

Transition level is obtained from:

- ☒ ATC or VOR ATIS.
- ☐ 1500 feet above airfield elevation.
- ☐ Jeppesen or Aerad flight guides.

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30

If an altimeter indicates 3500 FT with the actual QNH of 1004, 7 hPa set, the approximate pressure altitude is:

☒ 3745 FT

☐ 3255 FT

☐ 3500 FT

31

If the static vent became blocked during a descent, the altimeter would:

- ☐ read zero altitude
- ☒ under-read the correct altitude
- ☐ act normally
- ☐ over-read the correct altitude

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32

If the Pitot opening is blocked, which instruments would be affected (separate static vent)?

☐ ASI, Altimeter and VSI.

☒ ASI only.

☐ ASI and VSI.

33

The percentage error due to the positioning of the static vent is:

- ☒ either greater or smaller when comparing the ASI and the Machmeter,
- ☐ greater in the ASI than in the Machmeter
- ☐ the same in the Machmeter as in the ASI
- ☐ greater in the Machmeter than in the ASI

34

The rotor of the DGI spins up and away from the pilot when 090 is indicated. The latitude compensation nut situated on the near right hand side of the inner gimbal from the gyro axis, has been set to give zero drift on the ground at the equator. To compensate for earth rotation at 30 S the latitude compensating nut:

☐ must be adjusted inwards;

☐ must be adjusted outwards;

☒ is not adjusted since the latitude nut can only be used to correct for apparent wander in the northern hemisphere.

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35

The deviations of a direct reading magnetic compass effected only by coefficients B and C are, 3 E on 230 (C) and 4 W on 130 (C), therefore the deviation on heading 315 (C) is:

☒ 3,3 E

☐ 7,5 W

☐ 3,3 W

36

The principle of rigidity is used for the operation of the following gyroscopic instruments:

- ☒ Directional Gyro and Turn indicator.
- ☐ Artificial Horizon and Turn indicator.
- ☐ Directional Gyro and Artificial Horizon.

37

The angle of tilt of the rate gyro in a turn indicator is due to:

- ☐ the tension of the control spring opposing the angular tilt of the gyro;
- ☒ primary precession which is generated by an aeroplanes rate of turn.
- ☐ the force in the horizontal plane generated by secondary precession balancing the tilt caused by primary precession;

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38

What is the approximate angle of bank for a rate one turn at 110 knots?

☒ 25 degrees

☐ 18 degrees

☐ 30 degrees

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39

(Please view annex 022-10217A)

Considering the diagram n°1, the flight director command bars position indicates that you should:

- ☐ *bank your aircraft to the right and reduce the pitch angle.*
- ☒ *bank your aircraft to the left and increase the pitch angle.*
- ☐ *bank your aircraft to the left and reduce the pitch angle.*
- ☐ *bank your aircraft to the right and increase the pitch angle.*

See Attachments:

1. 022-10217a.gif

Annex

(Attachment
No.2)

40

The horizontal command bar of a flight director:

- 1 - repeats the position information given by the ILS in the horizontal plane
- 2 - repeats the position information given by the ILS in the vertical plane
- 3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

☐ 2.

☐ 1, 3.

☒ 3.

☐ 2, 3.

Attachment No. 1

Questions 21

figure 83

instrument_83.jpg



1



2



3



4

Figure 83. Altimeter/12,000 Feet © ASA

Attachment No. 2

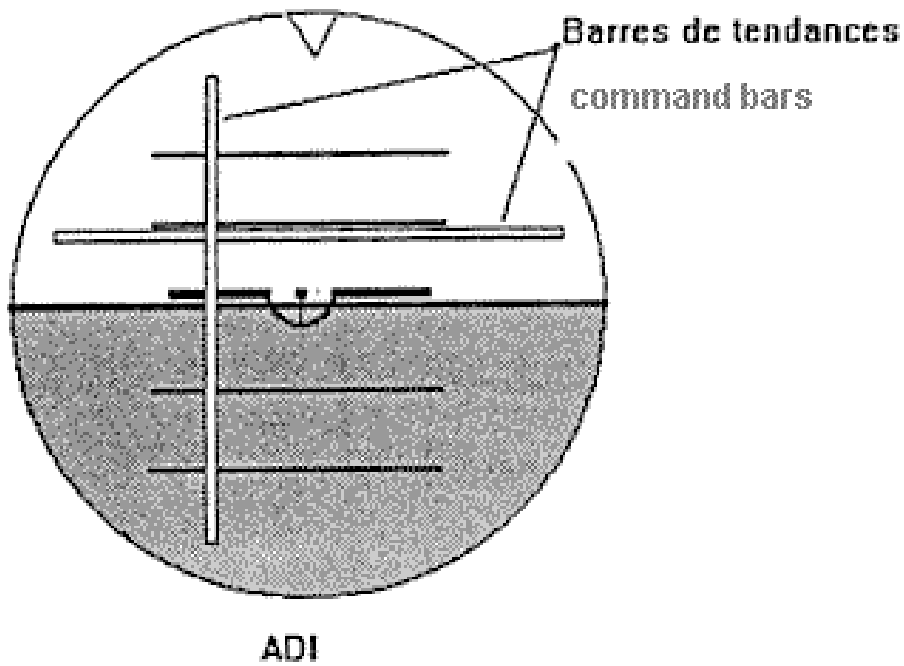
Questions 39

Annex

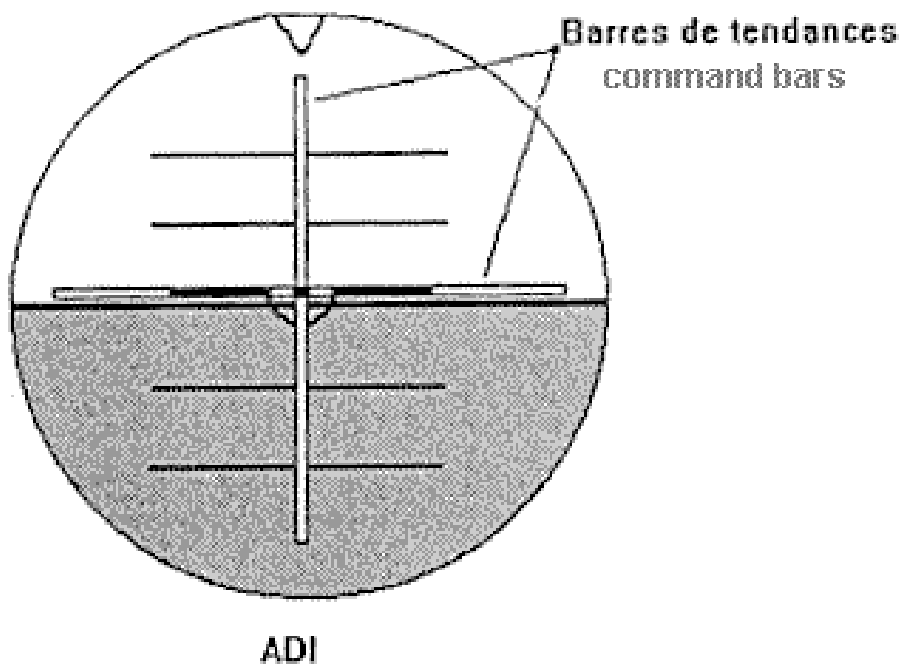
022-10217a.gif

1

022-10217 a



2



Comment No.2 for question 17:

Such a question is too lengthy to read, think and answer within a short time.